USN

RV COLLEGE OF ENGINEERING
Autonomous Institution affiliated to VTU
III Semester B.E. April -2024 Examinations
DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
NETWORK ANALYSIS AND CONTROL ENGINEERING

EC235AT

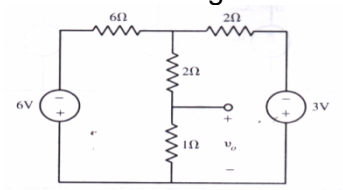
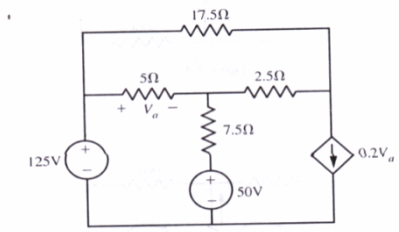
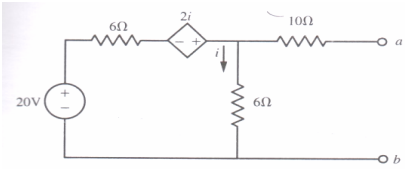
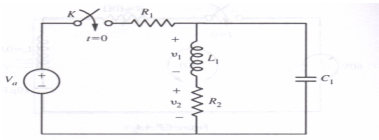
(2022 SCHEME)

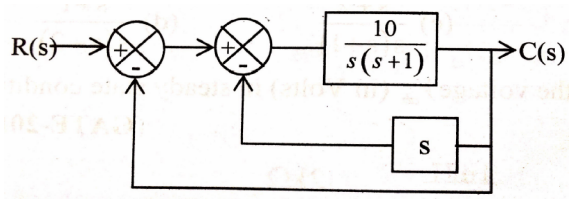
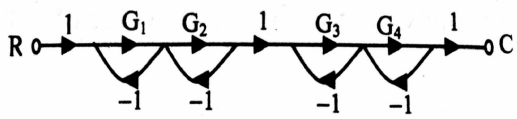
Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

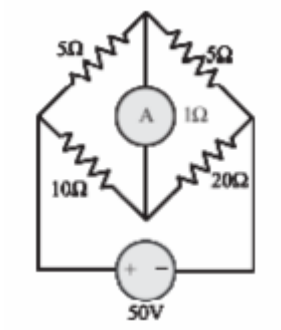
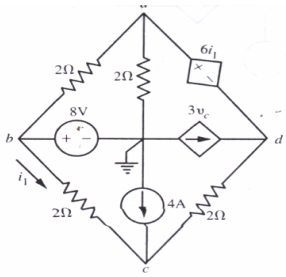
1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, and 9 and 10.

PART-A (Objective type for one or two marks)
(True & false and match the following questions are not permitted)

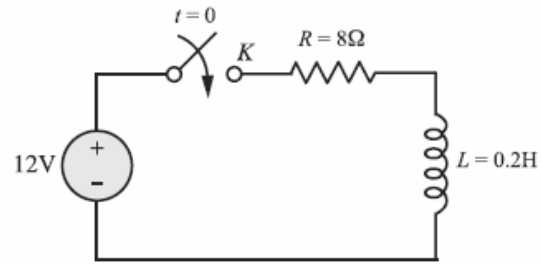
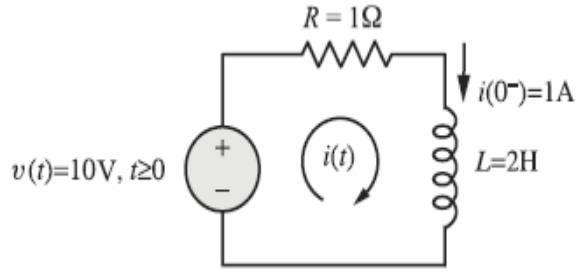
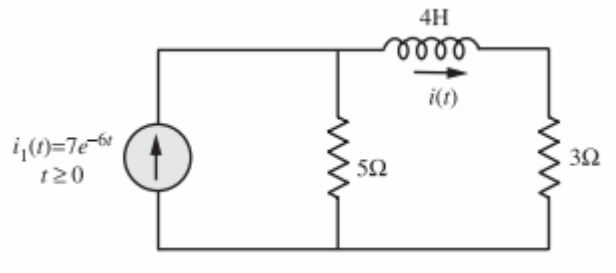
1	1.1	Use nodal analysis to find v_0 in the circuit in fig. below  Figure 1.60	2
	1.2	Find the total power delivered to the circuit shown below using mesh current method. 	2
	1.3	Find the thevinin equivalent for the circuit shown in fig. Below at terminals ab.  Figure 3.41	2
	1.4	In the network shown in fig. Below, switch K is closed at t=0 with zero capacitor voltage and zero inductor current. Find at $t=0^+$  Figure E.P. 4.6	2
	1.5	Initial & Final value theorem of Laplace transform is defined as _____	1
	1.6	Define transfer Function	1

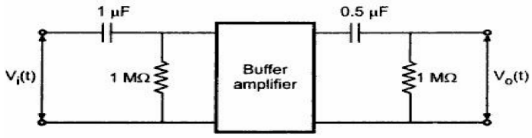
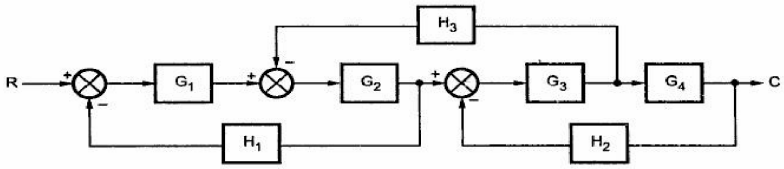
	1.7	For the system shown in figure the transfer function C(S)/R(S) is 	2
	1.8	C/R for the signal flow graph is 	2
	1.9	The Steady State Error for a stable TYPE 0 unity feedback system is_____	1
	1.10	Define Gain cross over Frequency and Phase Cross over Frequency.	1
	1.11	For G(S)H(S)=K/S(S+1)(S+4) determine the co-ordinates of valid breakaway points are_____	2
	1.12	For a second-order system with the closed-loop transfer function T(S)=9/S²+4S+9 the settling time for 2-percernt band, in seconds, is_____	2

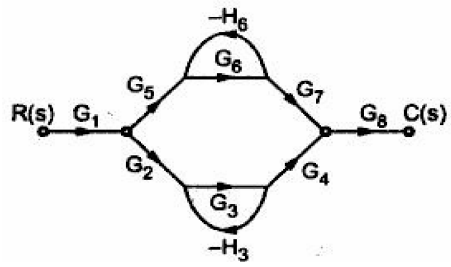
PART-B (Maximum subdivisions is limited to 4 in each question)

		UNIT-I	
2	a	Refer the circuit shown in Fig. Find current through the ammeter, and hence verify reciprocity theorem 	8
	b	Refer to the network shown below find the node voltages v_d and v_c. 	8

		UNIT-II	
3	a	In the given network switch K is closed at t=0 with zero current in the inductor. find the values of i, firat and second order rate of change of current with time at t=0⁺ if R=8 ohms and L=0.2H	8

			
	b	With mathematical equations explain the switching action of inductor and capacitor for initial and final conditions	8
		OR	
4	a	For the circuit shown find i(0) and i(∞) using initial and final value theorems 	8
	b	Find i(t) for the circuit of Fig.4b when $i_1(t) = 7e^{-6t}$ A for $t \geq 0$ and i(0) = 0. Also find i(∞).  Fig.4b	8

		UNIT-III	
5	a	Differentiate between Open Loop and Closed Loop Control systems.	6
	b	Determine the transfer function of the electrical system shown below. Assume Gain=1. 	10
		OR	
6	a	Determine the transfer function C(S)/R(S) of the system by block diagram reduction. 	8

	b	Using masons gain formula find C(S)/R(S) 	8

		UNIT-IV	
7	a	For a system G(S) H(S) =K/S²(S+2) (S+3). Find the value of K to limit Steady state error to 10, when input to the system is 1+10t+20t².	8
	b	The Open Loop transfer function of a unity feedback system is given by G(S)=K/S(ST+1) where K & T are positive constants. By What factor should the amplifier gain K be reduced so that the Peak overshoot of unit Step response of the system is reduced from 75% to 25%.	8
		OR	
8	a	Derive an expression for output response c(t) of underdamped 2nd order system for unit step input.	10
	b	For a unity feedback system having G(S)=15/(S+1)(S+3). Determine Characteristic equation, $\omega_n\omega_n$ &$\xi\xi$, Time at which first undershoot will occur, time period of Oscillations.	6

		UNIT-V	
9	a	Using Routh criteria determine the stability of a systems if Its loop transfer function has poles at s=0, s= -1, s= -3 and Zero at s= -5, gain K of forward path is 10.	8
	b	Sketch the root locus plot for all values of K ranging from 0 to infinity for a negative feedback control system characterized by $G(s)H(s) = \frac{K(s+1)}{s(s-1)(s^2+5s+20)}$	8
		OR	
10	a	Sketch the Bode plot for the system having G(S)H(S) =80/S(S+2)(S+20) and Determine GM, PM, ωgc, ωpc and comment on stability.	16

Signature of Scrutinizer: Signature of Chairman

Name: Name: